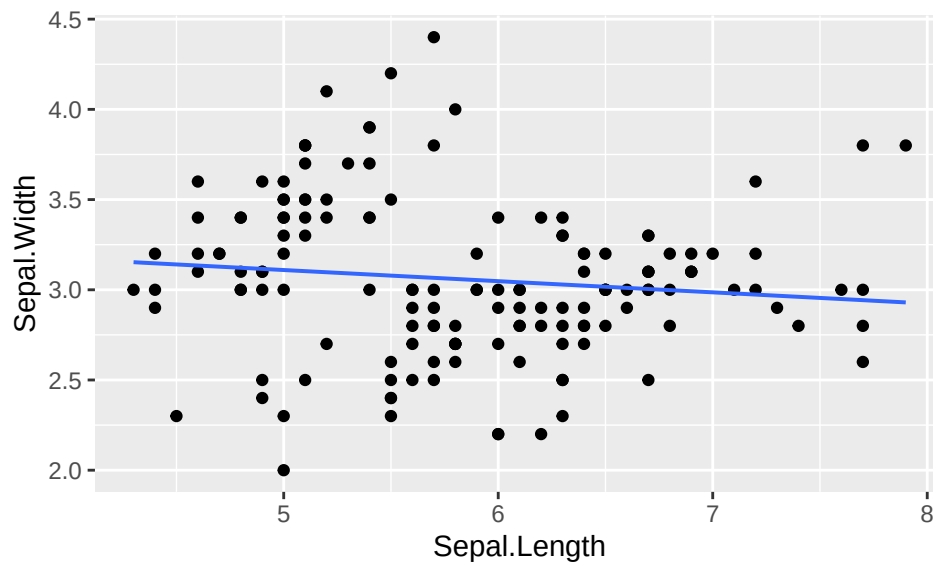


Testing for Parallelism of Regression Lines - Examples in R

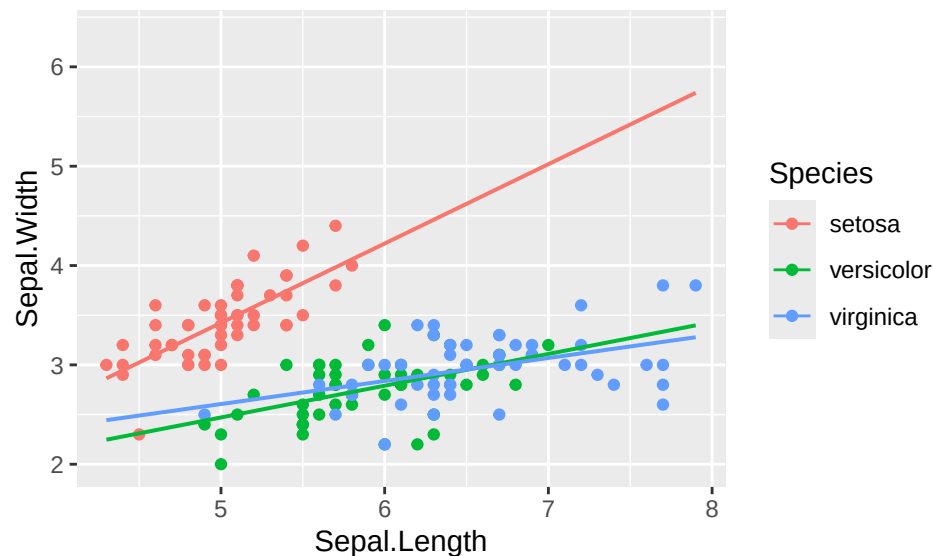
B.A. Hartlaub and A.S. Wagaman

Example - Iris

```
data(iris)
ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width)) +
  geom_point() +
  geom_lm()
```



```
ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width, color = Species)) +
  geom_point() +
  geom_lm()
```



In the iris data set, something interesting happens if we consider the plot of Sepal.Length versus Sepal.Width. The relationship looks negative if we ignore the three Species of iris present. Once you look by Species though, all three relationships are actually positive. Do these (parametric) regression lines have the same slope? Do the nonparametric equivalents have the same slope?

Parametric Test

We assume the conditions for inference are met. (Normality of errors and constant variance of errors - ask if you don't remember how to check these!)

```
fmbig <- lm(Sepal.Width ~ Sepal.Length*Species, data = iris)
msummary(fmbig) # so you can see what the output looks like
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.5694     0.5539  -1.028  0.305622
## Sepal.Length     0.7985     0.1104   7.235  2.55e-11 ***
## Speciesversicolor  1.4416     0.7130   2.022  0.045056 *
## Speciesvirginica  2.0157     0.6861   2.938  0.003848 **
## Sepal.Length:Speciesversicolor -0.4788     0.1337  -3.582  0.000465 ***
## Sepal.Length:Speciesvirginica  -0.5666     0.1262  -4.490  1.45e-05 ***
##
```

```
## Residual standard error: 0.2723 on 144 degrees of freedom
## Multiple R-squared:  0.6227, Adjusted R-squared:  0.6096
## F-statistic: 47.53 on 5 and 144 DF,  p-value: < 2.2e-16
```

```
fmsmall <- lm(Sepal.Width ~ Sepal.Length + Species, data = iris)
# use the anova command for test, smaller model first
anova(fmsmall, fmbig)
```

```
## Analysis of Variance Table
##
## Model 1: Sepal.Width ~ Sepal.Length + Species
## Model 2: Sepal.Width ~ Sepal.Length * Species
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     146 12.193
## 2     144 10.680  2     1.5132 10.201 7.19e-05 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Looking at the provided output, what do you conclude about whether or not you can use parallel slopes for these regression lines?

Discussion: The nested F test and the individual t tests for the coefficients of the interaction terms both suggest that three separate regression lines would be a better model than the model with parallel slopes.

What are the nonparametric slope estimates?

Good question. We use a new function called `rfit` from the `Rfit` package to obtain the nonparametric regression coefficients, including the interaction term.

```
npmod <- rfit(Sepal.Width ~ Sepal.Length*Species, data = iris)
summary(npmod, overall.test = 'drop') # not default
```

```
## Call:
## rfit.default(formula = Sepal.Width ~ Sepal.Length * Species,
##      data = iris)
##
## Coefficients:
##              Estimate Std. Error t.value   p.value
## (Intercept)    -0.31859    0.54049  -0.5895  0.556482
## Sepal.Length     0.74869    0.10772   6.9503 1.170e-10 ***
## Speciesversicolor  1.11573    0.69591   1.6033  0.111069
## Speciesvirginica  1.81013    0.66961   2.7033  0.007693 **
## Sepal.Length:Speciesversicolor -0.41536    0.13044  -3.1842  0.001779 **
## Sepal.Length:Speciesvirginica -0.52359    0.12316  -4.2511 3.805e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared (Robust): 0.5580881
## Reduction in Dispersion Test: 36.37136 p-value: 0
```

```
fmbig$coefficients
```

```
##              (Intercept)              Sepal.Length
##              -0.5694327              0.7985283
##              Speciesversicolor              Speciesvirginica
##              1.4415786              2.0157381
## Sepal.Length:Speciesversicolor Sepal.Length:Speciesvirginica
##              -0.4788090              -0.5666378
```

```
npmod$coefficients
```

```
##              (Intercept)              Sepal.Length
##              -0.3185916              0.7486922
##              Speciesversicolor              Speciesvirginica
##              1.1157271              1.8101285
## Sepal.Length:Speciesversicolor Sepal.Length:Speciesvirginica
##              -0.4153588              -0.5235886
```

The test is discussed in Hollander, Wolfe, and Chicken section 9.6.

How do the parametric and nonparametric model coefficients compare?

Discussion: The coefficients are slightly different, especially the estimated intercepts and the coefficients for `Speciesvirginica`.

```
npmodsmall <- rfit(Sepal.Width ~ Sepal.Length + Species, data = iris)
drop.test(npmod, npmodsmall) # smaller model goes second here
```

```
##
## Drop in Dispersion Test
## F-Statistic      p-value
## 8.89500550 0.00022781
```

If the same logic (as in the parametric case) applies here to the nonparametric case, what would you conclude about whether or not you can use parallel slopes for these regression lines?

Discussion: For our nonparametric models, the drop in dispersion test and the individual t tests for the coefficients of the interaction terms both suggest that three separate regression lines would be a better model than the model with parallel slopes.

Now let's try the Sen-Adichie procedure to see if they have equal slopes, and see if our logic holds.

Getting data into the form required for sen.adichie

For the sen.adichie function, the input is supposed to be a list of paired vectors, one per group. Each item in the list is a set of two paired vectors in the form of a matrix. The first column of each matrix is the x vector, the second is the y vector.

```
# grab variables in x, then y order
# forced use of dplyr select in case of conflicts
setxy <- filter(iris, Species == "setosa") %>%
  dplyr::select(Sepal.Length, Sepal.Width)
virxy <- filter(iris, Species == "virginica") %>%
  dplyr::select(Sepal.Length, Sepal.Width)
verxy <- filter(iris, Species == "versicolor") %>%
  dplyr::select(Sepal.Length, Sepal.Width)
```

Once you have these (the paired vectors), the sen.adichie function requires those to be put in a list.

```
z <- list(setxy, virxy, verxy) # it requires a list object
```

This should be the desired setup.

```
sen.adichie(z)
```

```
##
## Null: all slopes are equal
## V = 14.006, P = 0.001
```

Tada! Our test statistic V is 14.006, and the p-value is reported as 0.001. Let's confirm that we can get this p-value from the appropriate chi-square distribution. There are 3 groups so $k = 3$, and $k - 1 = 2$ for our df.

```
1 - pchisq(14.006, 2) # confirms p-value
```

```
## [1] 0.0009091504
```

```
qchisq(0.95, 2) # reject the null hypothesis if V is >= this value at alpha=0.05 level
```

```
## [1] 5.991465
```

So, what is our conclusion here?

Since 14.006 is greater than 5.991465, we reject the null hypothesis and conclude that the model with different slopes is better for the irises. The p-value is also tiny, which leads us to the same conclusion.

Data Sources

Iris data. Available in R with `data(iris)`. Original sources referenced in R help:

Fisher, R. A. (1936) The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7, Part II, 179–188. doi:10.1111/j.1469-1809.1936.tb02137.x.

Anderson, Edgar (1935). The irises of the Gaspé Peninsula, *Bulletin of the American Iris Society*, 59, 2–5.

Standard Reference

These materials were created for use in an undergraduate nonparametrics course. Where provided, textbook references refer to *Nonparametric Statistical Methods* (3rd Edition) by Hollander, Wolfe, and Chicken. Notation is chosen to be consistent with that text. However, materials may be used independently of the textbook.

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